

Solution Requirements

This document specifies all functional and non-functional requirements for OptiCrop v1.0, using MoSCoW prioritisation (Must Have, Should Have, Could Have, Won't Have).

Functional Requirements

ID	Requirement	Priority	Stakeholder	Ref
1	System shall accept N, P, K, temperature, humidity, pH, rainfall as inputs	Must Have	Farmer	PS-1
2	System shall validate all inputs for numeric type and acceptable range before prediction	Must Have	QA / Security	
3	System shall return a crop recommendation within 2 seconds of form submission	Must Have	Farmer	PS-1
4	System shall display the recommended crop name prominently on the result page	Must Have	Farmer	PS-1
5	System shall display a confidence score (%) alongside the recommendation	Should Have	Researcher	PS-2
6	System shall show the top-3 recommended crops ranked by confidence	Should Have	Agribusiness	PS-3
7	System shall provide a brief description of the recommended crop	Should Have	Farmer	PS-1
8	System shall allow users to clear and re-enter inputs without page refresh	Should Have	UX	
9	System shall log each prediction request (timestamp, inputs, output) to CSV	Could Have	Researcher	PS-2
10	System shall support English and Hindi language toggle	Won't Have (v1)	Roadmap	

Non-Functional Requirements

ID	Category	Requirement / Metric
1	Performance	Prediction response time < 2 seconds on local server
2	Availability	System uptime during growing season ≥ 99% (local deploy)
3	Usability	Task completion by first-time user ≥ 80% success in usability test

4	Portability	Runs on Windows, Linux, macOS verified via Python 3.10+ venv
5	Scalability	Handles up to 100 concurrent users (Gunicorn WSGI)
6	Accuracy	Model prediction accuracy on test set ϵ 99% (Random Forest)
7	Security	Input sanitisation against injection; no raw SQL or shell exec
8	Maintainability	Docstrings on all public functions; flake8 clean codebase

System Constraints & Assumptions

- " No external prediction API calls all inference must be local
- " Dataset: publicly available crop recommendation dataset (Kaggle, 2,200 rows)
- " Deployment environment: local server (no cloud infrastructure required for Phase 1)
- " Users have access to soil health card data or estimated NPK values from local labs
- " A modern browser (Chrome ϵ 90, Firefox ϵ 88) is available on the user's device